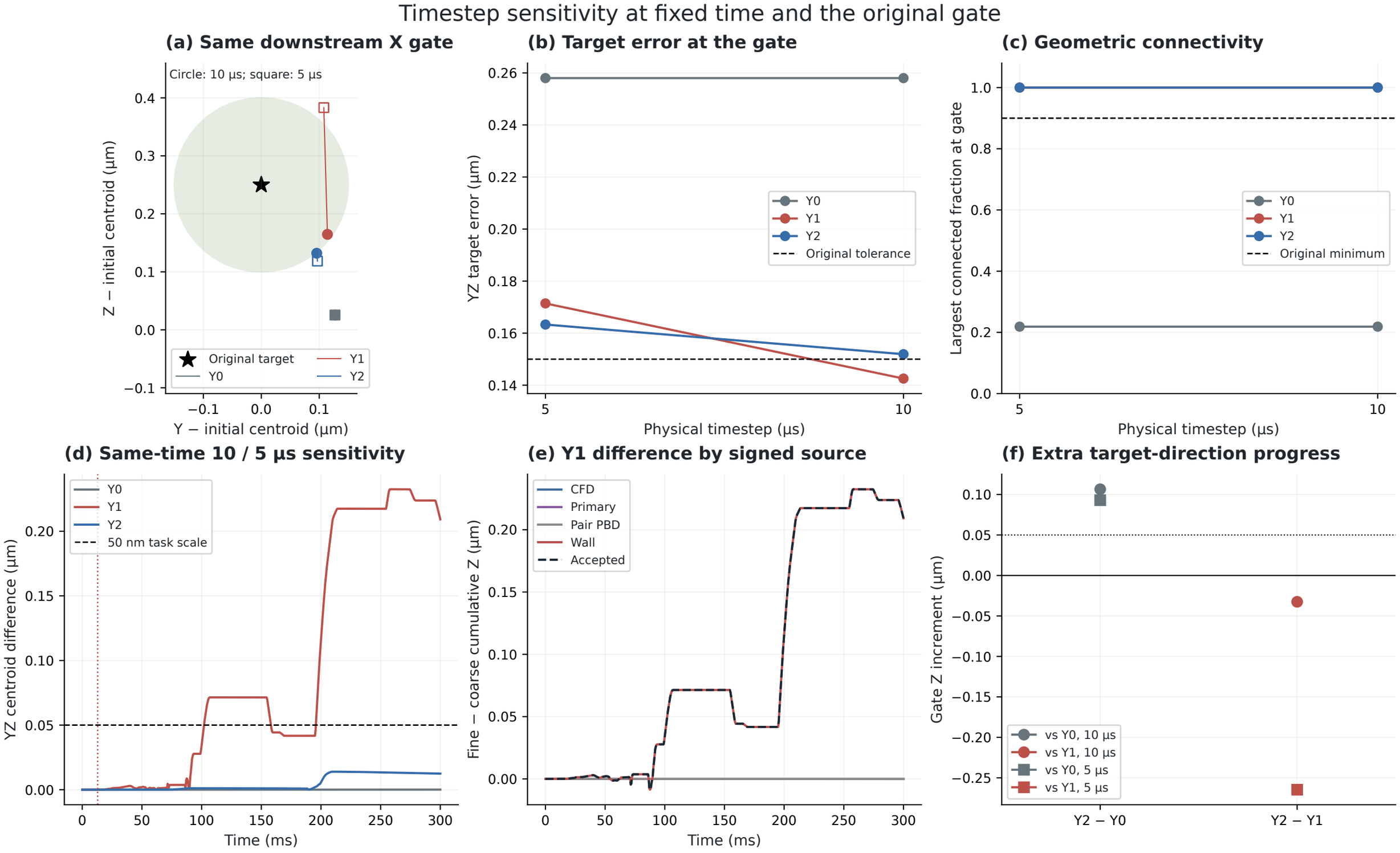
Timestep sensitivity at fixed time and the original gate
(a) Same downstream X gate
Circle: 10 µs; square: 5 µs
Z − initial centroid (µm)
Y − initial centroid (µm)
Original target
Y0
Y1
Y2
(b) Target error at the gate
YZ target error (µm)
Physical timestep (µs)
Y0
Y1
Y2
Original tolerance
(c) Geometric connectivity
Largest connected fraction at gate
Physical timestep (µs)
Y0
Y1
Y2
Original minimum
(d) Same-time 10 / 5 µs sensitivity
YZ centroid difference (µm)
Time (ms)
Y0
Y1
Y2
50 nm task scale
(e) Y1 difference by signed source
Fine − coarse cumulative Z (µm)
Time (ms)
CFD
Primary
Pair PBD
Wall
Accepted
(f) Extra target-direction progress
Gate Z increment (µm)
vs Y0, 10 µs
vs Y1, 10 µs
vs Y0, 5 µs
vs Y1, 5 µs
Y2 − Y0
Y2 − Y1